CAMBRIDGE IGCSE™ & O LEVEL COMPUTER SCIENCE (0478 / 2210)

Chapter 1: Data Representation • Comprehensive Master Revision Guide

TOPICS 1.1: NUMBER SYSTEMS & 1.2: TEXT, SOUND AND IMAGES

TOPIC 1.1: NUMBER SYSTEMS

1.1.1 Binary, Denary, and Hexadecimal Systems

• **Binary (Base 2):** Uses 2 digits (`0` and `1`). Place values: $128, 64, 32, 16, 8, 4, 2, 1$.
• **Denary (Base 10):** Uses 10 digits (0 to 9). Standard human counting system.
• **Hexadecimal (Base 16):** Uses 16 symbols (0–9 and $ ext{A}=10, ext{B}=11, ext{C}=12, ext{D}=13, ext{E}=14, ext{F}=15$).
• **Conversion Rule:** 1 Hexadecimal digit represents exactly **4 binary bits (1 nibble)**.

1.1.2 Why Hexadecimal is Used in Computer Science

1. **More compact:** Much shorter representation than binary (takes up less screen/memory space).
2. **Human readability:** Much easier for humans to read, understand, and remember.
3. **Fewer errors:** Greatly reduces human transcription/typing mistakes when debugging.
4. **Modern Applications:** MAC Addresses (e.g. `00:1A:2B:3C:4D:5E`), IPv6 Addresses, HTML/RGB Colour Codes (e.g. `#FF5733`), Memory Dumps, and Assembly opcodes.

1.1.3 Binary Addition & Overflow Error

• **Addition Rules:** $0+0=0$, $0+1=1$, $1+1=10_2$ (0 carry 1), $1+1+1=11_2$ (1 carry 1).
• **Overflow Error:** Occurs when the sum of two binary numbers exceeds the maximum value that the register can store (e.g., >255 in an 8-bit register). An extra 9th carry bit is generated from the Most Significant Bit (MSB), causing data truncation and inaccurate calculations.

1.1.4 Logical Shifts (Left & Right)

• **Logical Left Shift (n places):** Shifts all bits to the left, inserting 0s on the right. *Arithmetic effect:* Multiplies the denary value by 2^n. If a `1` is shifted out of the MSB, an **overflow error** occurs.
• **Logical Right Shift (n places):** Shifts all bits to the right, inserting 0s on the left. *Arithmetic effect:* Divides the denary value by 2^n (integer division). If a `1` is shifted out of the LSB, **loss of precision/fractional data** occurs.

1.1.5 Two's Complement (Signed Binary Numbers)

• Used to represent both positive and negative integers in binary.
• In an 8-bit register, the MSB has a negative place value of **−128** (Range: **−128 to +127**).
• **Method to convert negative denary (e.g. −45):**
  1. Write +45 in 8-bit binary: `00101101`
  2. Invert all bits (One's complement): `11010010`
  3. Add 1 to the result: `11010010 + 1 = 11010011` (Two's complement).

TOPIC 1.2: TEXT, SOUND AND IMAGES

1.2.1 Text Representation (Character Sets)

• **Character Set:** A defined list of characters recognized and encoded into binary numbers by a computer system.
• **ASCII (American Standard Code for Information Interchange):** Uses 7 bits (128 characters) or 8 bits (Extended ASCII, 256 characters). Only supports English letters, digits, and control codes.
• **Unicode:** Uses 16 or 32 bits (UTF-8, UTF-16). Can represent over 1,000,000 unique characters, supporting all global languages (Bengali, Chinese, Arabic, etc.), scientific symbols, and emojis.

1.2.2 Sound Representation (Sampling)

• Continuous analogue sound waves are converted into digital binary data using an **ADC (Analogue-to-Digital Converter)** by taking sound samples at regular time intervals.
• **Sample Rate:** Number of sound samples captured per second (measured in **Hz / kHz**).
• **Sample Resolution (Bit Depth):** Number of bits used to store each audio sample.
• **Trade-off:** Increasing Sample Rate or Bit Depth improves **sound quality** and accuracy, but increases the **file size** and bandwidth requirements.

Sound File Size (bits) = Sample Rate (Hz) × Sample Resolution (bits) × Duration (seconds) × Channels

1.2.3 Image Representation (Bitmap Images)

• A **bitmap image** is composed of a grid of tiny dots called **pixels (picture elements)**.
• **Image Resolution:** The dimensions of the image in pixels ($ ext{Width} imes ext{Height}$, e.g. $1920 imes 1080$).
• **Colour Depth:** The number of bits used to represent the colour of each pixel (n bits allows 2^n unique colours, e.g. 24-bit True Colour allows $2^{24} = 16,777,216$ colours).
• **Metadata:** Additional information stored with the image file (dimensions, colour depth, date created, camera details).

Image File Size (bits) = Image Width (px) × Image Height (px) × Colour Depth (bits)

1.2.4 Data Compression (Lossy, Lossless & RLE)

Feature	Lossy Compression	Lossless Compression
Definition	Permanently removes redundant/unnoticeable data.	Reduces file size without deleting any original data.
Decompression	Original file cannot be recreated exactly.	Original file is reconstructed bit-for-bit with zero loss.
Compression Ratio	Very high reduction in file size.	Moderate reduction in file size.
File Examples	**JPEG, MP3, MP4**	**PNG, GIF, FLAC, ZIP, SVG**

• **Run-Length Encoding (RLE):** A lossless algorithm that replaces consecutive repeated data values (runs) with a pair: `[count, value]` (e.g. `WWWWWB B B → 5W 3B`).